MA087G Mathematical Statistics
Full Solution Pass Guide
2026-01-07

Source notes before solving

The official exam PDF was image-only: no embedded text could be extracted, but the rendered pages were readable.

Two details are worth knowing before using this as a study guide:

- Uppgift 2 is internally inconsistent. The displayed density is $f(x) = ax^2(2-x)$, but the printed fallback comment says $f(x) = 5x^3(2-x)/8$. I solve the displayed density, since that is the actual formula in the question, and I also give the fallback answers.
- Uppgift 4b does not explicitly state that X and Y are independent. To compute the requested probability from the given information, independence has to be assumed. I use that standard exam assumption.

Uppgift 1

What the question wants

This is a probability table problem with independence and conditional probability. The fastest move is to fill in $P(A)$, $P(B)$, and $P(A \cap B)$.

Fast exam method

Translate the Swedish words into set probabilities:

$$P(A^c \cap B^c) = 0.60, \qquad P(A \cap B) = 0.20, \qquad P(A^c \cap B) = 0.10.$$

Then use $P(A \cup B) = 1 - P(A^c \cap B^c)$.

Solution

First,

$$P(A \cup B) = 1 - 0.60 = 0.40.$$

Also,

$$P(B) = P(A \cap B) + P(A^c \cap B) = 0.20 + 0.10 = 0.30.$$

Use the union formula:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ 0.40 &= P(A) + 0.30 - 0.20 \\ P(A) &= 0.30. \end{aligned}$$

For independence, check whether

$$P(A \cap B) = P(A)P(B).$$

Here

$$P(A)P(B) = 0.30 \cdot 0.30 = 0.09 \neq 0.20.$$

So A and B are not independent.

For the conditional probability:

$$P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)} = \frac{0.10}{1 - 0.30} = \frac{0.10}{0.70} = \frac{1}{7}.$$

Answer

$$\boxed{A \text{ and } B \text{ are not independent}}, \qquad \boxed{P(B \mid A^c) = \frac{1}{7} \approx 0.143.}$$

Exam trick

For independence, do not compare $P(A \cap B)$ with $P(A) + P(B)$. Always compare $P(A \cap B)$ with $P(A)P(B)$.

Repeated exam pattern

No earlier matching pattern found yet. Track this as a new pattern: conditional probability table with independence check.

Uppgift 2

What the question wants

This is a continuous random variable problem. You normalize a density, build the distribution function, then calculate probability, expected value, and standard deviation.

Fast exam method

For a density:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

For a CDF:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

For moments:

$$E(X) = \int x f(x)\,dx, \qquad E(X^2) = \int x^2 f(x)\,dx, \qquad \operatorname{Var}(X) = E(X^2) - [E(X)]^2.$$

Solution

Using the displayed density in the exam:

$$f(x) = \begin{cases} ax^2(2-x), & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Normalize:

$$\begin{aligned} 1 &= \int_0^2 ax^2(2-x)\,dx \\ &= a\int_0^2 (2x^2 - x^3)\,dx \\ &= a\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 \\ &= a\left(\frac{16}{3} - 4\right) = a \cdot \frac{4}{3}. \end{aligned}$$

So

$$a = \frac{3}{4}.$$

The distribution function is

$$F(x) = \begin{cases} 0, & x < 0, \\ \displaystyle\int_0^x \frac{3}{4}t^2(2-t)\,dt, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

For $0 \le x \le 2$:

$$\begin{aligned} F(x) &= \frac{3}{4}\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^x \\ &= \frac{x^3}{2} - \frac{3x^4}{16}. \end{aligned}$$

Thus

$$F(x) = \begin{cases} 0, & x < 0, \\ \dfrac{x^3}{2} - \dfrac{3x^4}{16}, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

Now calculate $P(X > 1)$:

$$\begin{aligned} P(X > 1) &= 1 - P(X \le 1) \\ &= 1 - F(1) \\ &= 1 - \left(\frac{1}{2} - \frac{3}{16}\right) \\ &= 1 - \frac{5}{16} = \frac{11}{16}. \end{aligned}$$

Expected value:

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{4}x^2(2-x)\,dx \\ &= \frac{3}{4}\int_0^2 (2x^3 - x^4)\,dx \\ &= \frac{3}{4}\left[\frac{x^4}{2} - \frac{x^5}{5}\right]_0^2 \\ &= \frac{3}{4}\left(8 - \frac{32}{5}\right) = \frac{6}{5}. \end{aligned}$$

Second moment:

$$\begin{aligned} E(X^2) &= \int_0^2 x^2 \cdot \frac{3}{4}x^2(2-x)\,dx \\ &= \frac{3}{4}\int_0^2 (2x^4 - x^5)\,dx \\ &= \frac{3}{4}\left[\frac{2x^5}{5} - \frac{x^6}{6}\right]_0^2 \\ &= \frac{3}{4}\left(\frac{64}{5} - \frac{32}{3}\right) = \frac{8}{5}. \end{aligned}$$

Therefore

$$\begin{aligned} \mathrm{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{8}{5} - \left(\frac{6}{5}\right)^2 \\ &= \frac{4}{25}, \end{aligned}$$

and

$$\mathrm{SD}(X) = \sqrt{\frac{4}{25}} = \frac{2}{5}.$$

Answer

For the displayed density $ax^2(2-x)$:

$$\boxed{a = \frac{3}{4}}$$

$$\boxed{F(x) = \begin{cases} 0, & x < 0, \\ \frac{x^3}{2} - \frac{3x^4}{16}, & 0 \le x \le 2, \\ 1, & x > 2, \end{cases}}$$

$$\boxed{P(X > 1) = \frac{11}{16}}, \qquad \boxed{E(X) = \frac{6}{5}}, \qquad \boxed{\mathrm{SD}(X) = \frac{2}{5}.}$$

If the printed fallback density $5x^3(2-x)/8$ was intended instead, then:

$$F(x) = \frac{5x^4}{16} - \frac{x^5}{8}, \quad P(X > 1) = \frac{13}{16}, \quad E(X) = \frac{4}{3}, \quad \mathrm{SD}(X) = \sqrt{\frac{8}{63}}.$$

Exam trick

When a density is only nonzero on an interval, every integral should run over that interval, here 0 to 2. For $P(X > 1)$, using $1 - F(1)$ is fastest.

Repeated exam pattern

No earlier matching pattern found yet. Track this as a new pattern: continuous density normalization plus CDF and moments.

Uppgift 3

What the question wants

This is a minimum-of-independent-exponentials problem. The key shortcut is to use survival probabilities.

Fast exam method

If $Z = \min(X, Y)$, then

$$P(Z > z) = P(X > z \text{ and } Y > z).$$

For independent exponential variables, the rates add.

Solution

Given

$$X \sim \text{Exp}(0.04), \qquad Y \sim \text{Exp}(0.05),$$

with X and Y independent.

For $z \geq 0$:

$$\begin{aligned} P(Z > z) &= P(\min(X, Y) > z) \\ &= P(X > z,\ Y > z) \\ &= P(X > z)P(Y > z) \\ &= e^{-0.04z}e^{-0.05z} \\ &= e^{-0.09z}. \end{aligned}$$

So

$$Z \sim \text{Exp}(0.09).$$

Equivalently,

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ 1 - e^{-0.09z}, & z \geq 0. \end{cases}$$

Now

$$P(Z \geq 10) = P(Z > 10) = e^{-0.09 \cdot 10} = e^{-0.9} \approx 0.407.$$

For an exponential variable with rate λ,

$$E(Z) = \frac{1}{\lambda}.$$

Thus

$$E(Z) = \frac{1}{0.09} = \frac{100}{9} \approx 11.11.$$

Answer

$$\boxed{Z \sim \text{Exp}(0.09)}, \qquad \boxed{P(Z \geq 10) = e^{-0.9} \approx 0.407}, \qquad \boxed{E(Z) = \frac{100}{9} \approx 11.11.}$$

Exam trick

Minimum means both variables must survive past z. Maximum would need at least one to survive, which is a different calculation.

Repeated exam pattern

No earlier matching pattern found yet. Track this as a new pattern: minimum of independent exponential variables.

Uppgift 4

What the question wants

Part a is a discrete distribution with expected value and standard deviation. Part b is a linear combination of normal variables.

Fast exam method

For part a, turn the words into probabilities. For part b, write the cost as a random variable and use that a linear combination of independent normal variables is normal.

Solution

Part a. The possible contributions are 0, 50, and 100 kr.

We are told:

$$P(X = 0) = 0.46.$$

The remaining probability is

$$1 - 0.46 = 0.54.$$

Let

$$P(X = 100) = q.$$

The 50 kr contribution is twice as common:

$$P(X = 50) = 2q.$$

So

$$q + 2q = 0.54 \quad \Rightarrow \quad 3q = 0.54 \quad \Rightarrow \quad q = 0.18.$$

Therefore

$$P(X = 100) = 0.18, \qquad P(X = 50) = 0.36, \qquad P(X = 0) = 0.46.$$

Expected value:

$$\begin{aligned} E(X) &= 0 \cdot 0.46 + 50 \cdot 0.36 + 100 \cdot 0.18 \\ &= 18 + 18 = 36. \end{aligned}$$

Second moment:

$$\begin{aligned} E(X^2) &= 0^2 \cdot 0.46 + 50^2 \cdot 0.36 + 100^2 \cdot 0.18 \\ &= 2500 \cdot 0.36 + 10000 \cdot 0.18 \\ &= 900 + 1800 = 2700. \end{aligned}$$

Variance and standard deviation:

$$\begin{aligned} \mathrm{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= 2700 - 36^2 \\ &= 2700 - 1296 = 1404, \\ \mathrm{SD}(X) &= \sqrt{1404} \approx 37.47. \end{aligned}$$

Part b. The trip cost is

$$C = 57 + 14X + 9Y.$$

Using the standard convention $N(\mu, \sigma)$, we have

$$X \sim N(3, 1), \qquad Y \sim N(8, 2).$$

Assuming X and Y are independent:

$$E(C) = 57 + 14 \cdot 3 + 9 \cdot 8 = 171.$$

The variance is

$$\begin{aligned} \mathrm{Var}(C) &= 14^2 \,\mathrm{Var}(X) + 9^2 \,\mathrm{Var}(Y) \\ &= 14^2 \cdot 1^2 + 9^2 \cdot 2^2 \\ &= 196 + 324 = 520. \end{aligned}$$

So

$$C \sim N(171, \sqrt{520}).$$

Now standardize:

$$\begin{aligned} P(150 \le C \le 200) &= P\left(\frac{150 - 171}{\sqrt{520}} \le Z \le \frac{200 - 171}{\sqrt{520}}\right) \\ &= P(-0.921 \le Z \le 1.272) \\ &= \Phi(1.272) - \Phi(-0.921) \\ &\approx 0.720. \end{aligned}$$

Answer

x	0	50	100
$P(X = x)$	0.46	0.36	0.18

$\boxed{E(X) = 36 \text{ kr}}$, $\boxed{\mathrm{SD}(X) \approx 37.47 \text{ kr}}$.

$\boxed{P(150 \le C \le 200) \approx 0.720.}$

Exam trick

For linear normal questions, do the mean and variance separately. Means add directly, but standard deviations do not: variances add.

Repeated exam pattern

No earlier matching pattern found yet. Track this as two new patterns: discrete distribution with moments, and linear combination of normal variables.

Uppgift 5

What the question wants

This is a maximum likelihood problem for a geometric-type distribution. X counts successes before the first miss.

Fast exam method

Write the likelihood, take log, differentiate, set equal to zero. The repeated trick is that products become sums in the log-likelihood.

Solution

The probability function is

$$p_X(k) = p^k(1-p), \qquad k = 0, 1, 2, \ldots$$

For observations $x_1, \ldots, x_n$, the likelihood is

$$\begin{aligned} L(p) &= \prod_{i=1}^{n} p^{x_i}(1-p) \\ &= p^{\sum_{i=1}^{n} x_i}(1-p)^n. \end{aligned}$$

Let

$$S = \sum_{i=1}^{n} x_i.$$

Then

$$L(p) = p^S(1-p)^n.$$

The log-likelihood is

$$\ell(p) = S \ln p + n \ln(1-p).$$

Differentiate:

$$\ell'(p) = \frac{S}{p} - \frac{n}{1-p}.$$

Set $\ell'(p) = 0$:

$$\begin{aligned} \frac{S}{p} &= \frac{n}{1-p} \\ S(1-p) &= np \\ S &= p(S+n) \\ \hat{p}_{\mathrm{ML}} &= \frac{S}{S+n}. \end{aligned}$$

Since $S = n\bar{x}$, this can also be written as

$$\hat{p}_{\mathrm{ML}} = \frac{\bar{x}}{\bar{x}+1}.$$

For the data

$$2,\ 0,\ 9,\ 1,\ 6,\ 7,\ 5,\ 3,$$

we have

$$S = 2+0+9+1+6+7+5+3 = 33, \qquad n = 8.$$

So

$$\hat{p}_{\mathrm{ML}} = \frac{33}{33+8} = \frac{33}{41} \approx 0.805.$$

Answer

$$\boxed{\hat{p}_{\mathrm{ML}} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} x_i + n} = \frac{\bar{x}}{\bar{x}+1}}$$

and for the observed data:

$$\boxed{\hat{p}_{\mathrm{ML}} = \frac{33}{41} \approx 0.805.}$$

Exam trick

This geometric version counts successes before the first failure, so the mean is $\frac{p}{1-p}$. That also hints that the estimate should look like $\bar{x}/(\bar{x}+1)$.

Repeated exam pattern

No earlier matching pattern found yet. Track this as a new pattern: ML estimation for geometric distribution.

Uppgift 6

What the question wants

This is a confidence interval and hypothesis test for a normal mean when the population standard deviation is unknown.

Fast exam method

Use a t-interval because the sample standard deviation s is given:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

For the hypothesis test, use the same standard error:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Solution

Given:

$$n = 30, \qquad \bar{x} = 1260, \qquad s = 150.$$

Part a. A 90% confidence interval has $\alpha = 0.10$, so $\alpha/2 = 0.05$. With $n-1 = 29$ degrees of freedom:

$$t_{0.95,29} \approx 1.699.$$

The standard error is

$$\frac{s}{\sqrt{n}} = \frac{150}{\sqrt{30}} \approx 27.386.$$

Therefore:

$$\begin{aligned}\bar{x} \pm t_{0.95,29}\frac{s}{\sqrt{n}} &= 1260 \pm 1.699 \cdot 27.386 \\ &= 1260 \pm 46.53.\end{aligned}$$

So the interval is

$$(1213.47,\ 1306.53).$$

Part b. The consultant claimed the mean daily income will be 1280 kr. Since the question asks whether that claim seems reasonable or can be rejected, use a two-sided test:

$$H_0 : \mu = 1280, \qquad H_1 : \mu \neq 1280.$$

Part c. The test statistic is

$$\begin{aligned}T &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ &= \frac{1260 - 1280}{150/\sqrt{30}} \\ &\approx -0.730.\end{aligned}$$

At significance level 10%, with 29 degrees of freedom, the two-sided critical values are

$$\pm t_{0.95,29} \approx \pm 1.699.$$

Since

$$-1.699 < -0.730 < 1.699,$$

we do not reject H_0.

Same conclusion using the confidence interval: 1280 lies inside $(1213.47, 1306.53)$, so the consultant's value is plausible at the 10% level.

Answer

$$\boxed{90\%\text{ CI for }\mu:\ (1213.47,\ 1306.53)\text{ kr}}$$

$$\boxed{H_0 : \mu = 1280, \qquad H_1 : \mu \neq 1280}$$

$$\boxed{\text{Do not reject } H_0\text{. The consultant's claim is not rejected.}}$$

Exam trick

If σ is unknown and you are given s, think t-distribution. A hypothesis value inside the matching confidence interval is not rejected.

Repeated exam pattern

No earlier matching pattern found yet. Track this as two new patterns: confidence interval for mean with unknown σ, and two-sided hypothesis test for mean.

Mini-cheatsheet from this exam

Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A, B \text{ independent} \iff P(A \cap B) = P(A)P(B)$$

$$P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)}$$

Continuous distributions

Density check:

$$\int f(x)\,dx = 1.$$

CDF:

$$F(x) = \int_{-\infty}^{x} f(t)\,dt.$$

Moments:

$$E(X) = \int x f(x)\,dx, \qquad \mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Exponential minimum

If $X \sim \mathrm{Exp}(\lambda_1)$, $Y \sim \mathrm{Exp}(\lambda_2)$, independent, and $Z = \min(X, Y)$, then

$$Z \sim \mathrm{Exp}(\lambda_1 + \lambda_2).$$

Discrete variables

$$E(X) = \sum x P(X = x), \qquad E(X^2) = \sum x^2 P(X = x), \qquad \mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Normal linear combinations

If $C = a + bX + cY$ and X, Y are independent normal variables, then C is normal:

$$E(C) = a + bE(X) + cE(Y),$$

$$\operatorname{Var}(C) = b^2 \operatorname{Var}(X) + c^2 \operatorname{Var}(Y).$$

Maximum likelihood

For

$$p_X(k) = p^k(1-p),$$

with data $x_1, \ldots, x_n$:

$$\hat{p}_{\mathrm{ML}} = \frac{\sum x_i}{\sum x_i + n}.$$

Confidence intervals and tests

Mean, unknown σ:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

Test statistic:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

At 10% two-sided significance, compare with $\pm t_{0.95, n-1}$.